

1.

In this question you must show all stages of your working.
Solutions based entirely on calculator technology are not acceptable.

- (i) (a) Use the Euclidean algorithm to determine the highest common factor h of 105 and 24

(3)

- (b) Hence determine integers a and b such that

$$105a + 24b = h$$

(3)

- (ii) Determine the remainder when 179^5 is divided by 11

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



2. (i) Using a suitable algorithm and without performing any division, determine whether 13 306 617 is divisible by 9

(2)

- (ii) The group $G = \{1, 3, 7, 9, 11, 13, 17, 19\}$ has multiplication modulo 20 as its operation.

- (a) Complete the following Cayley table for G

$\times_{20}$	1	3	7	9	11	13	17	19
1	1	3	7	9	11	13	17	19
3	3		1			19	11	
7	7	1			17			13
9	9		3	1		17	13	
11	11			19	1			
13	13		11			9	1	7
17	17	11						3
19	19		13		9		3	

A copy of this table is given on page 9 if you need to rewrite your Cayley table.

(3)

- (b) State the inverse of the element 7

(1)

- (c) Determine the order of the element 13

(1)

- (d) Write down a subgroup of G of order 4

(1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Only use this grid if you need to rewrite your Cayley table.

$\times_{20}$	1	3	7	9	11	13	17	19
1	1	3	7	9	11	13	17	19
3	3		1			19	11	
7	7	1			17			13
9	9		3	1		17	13	
11	11			19	1			
13	13		11			9	1	7
17	17	11						3
19	19		13		9		3	

(Total for Question 2 is 8 marks)



- 3.** A loan of £180 000 is taken out to buy a house.

The monthly interest rate on the loan is 0.15%

The interest is added to the balance of the loan at the end of each month.

To repay the loan, £900 is repaid at the end of each month, immediately after the interest has been added.

Let B_n thousands of pounds be the balance of the loan at the end of month n after the interest has been added and the £900 repaid.

- (a) Explain, in the context of the problem, why the balance of the loan, B_n , can be modelled by the recurrence relation

$$B_n = 1.0015B_{n-1} - 0.9 \quad B_0 = 180 \quad n \in \mathbb{Z}^+ \quad (2)$$

- (b) State an assumption that must be made for this model to be valid. (1)
- (c) Solve the recurrence relation to determine a closed form for B_n (5)
- (d) Hence determine the time it will take to repay the loan. Give your answer in years and months to the nearest month. (2)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)



4.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

$$\mathbf{A} = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$$

Determine a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$$

(7)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 7 marks)



- 5.** In an Argand diagram, the curve C with equation

$$\arg\left(\frac{z+4}{z-2i}\right) = \frac{\pi}{4}$$

represents an arc of a circle.

Given that $z = x + iy$, where x and y are real numbers,

- (a) show that this circle has equation

$$x^2 + y^2 + ax + by + c = 0$$

where a , b and c are constants to be determined.

(4)

- (b) For the curve C , determine the exact minimum value of $|z|$

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 7 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/22**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
22: Further Pure Mathematics 2
(Part of option A only)

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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- The total mark for this part of the examination is 40. There are 5 questions.
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Advice

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Turn over ►

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1. (i) The table below is a Cayley table for the group G with operation $\circ$

$\circ$	a	b	c	d	e	f
a	d	c	b	a	f	e
b	e	f	a	b	c	d
c	f	e	d	c	b	a
d	a	b	c	d	e	f
e	b	a	f	e	d	c
f	c	d	e	f	a	b

- (a) State which element is the identity of the group.

(1)

- (b) Determine the inverse of the element $(b \circ c)$

(2)

- (c) Give a reason why the set $\{a, b, e, f\}$ cannot be a subgroup of G . You must justify your answer.

(1)

- (d) Show that the set $\{b, d, f\}$ is a subgroup of G .

(2)

- (ii) Given that H is a group with an element x of order 3 and an element y of order 6 satisfying

$$yx = xy^5$$

show that $y^3xy^3x^2$ is the identity element.

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. Tiles are sold in boxes with 21 tiles in each box.

The tiles are laid out in x rows of 5 tiles and y rows of 6 tiles.

All the tiles from a box are used before the next box is opened.

When all the rows of tiles have been laid, there are n tiles left in the last opened box.

- (a) Write down a congruence expression for n in the form

$$ax + by \pmod{c}$$

where a , b and c are integers.

(1)

Given that

- exactly 43 rows of tiles are laid
 - there are no tiles left in the last opened box
- (b) use your congruence expression to determine the minimum number of rows of 6 tiles laid.

(5)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$\mathbf{A} = \begin{pmatrix} 3 & k \\ -5 & 2 \end{pmatrix}$$

where k is a constant.

Given that there exists a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is a diagonal matrix where

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 8 & 0 \\ 0 & -3 \end{pmatrix}$$

(a) show that $k = -6$

(3)

(b) determine a suitable matrix $\mathbf{P}$

(4)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



- $$|z - (-3 + 3i)| = \alpha |z - (1 + 3i)|$$

(a) sketch, on an Argand diagram, the circle C **(2)**

(b) explain why the value of α is 3 **(1)**

$$R = \left\{ z \in \mathbb{C} : \beta \leq \arg z \leq \frac{\pi}{2} \right\}$$

- (c) Determine the maximum value of β
- Give your answer in radians to 3 significant figures.
- (6)**

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5.

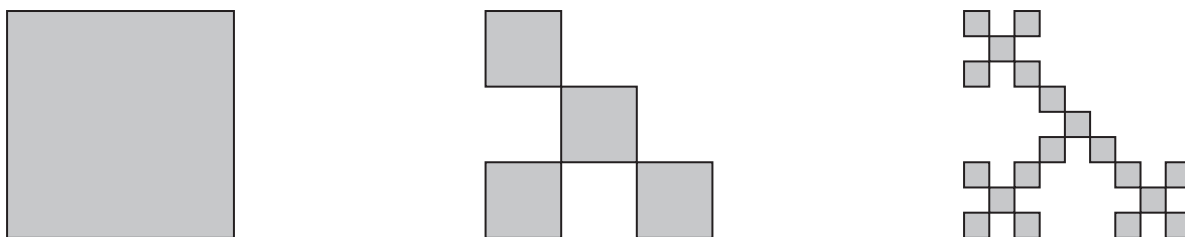


Figure 1

Figure 1 shows the first three stages of a pattern that is created by a recursive process.

The process starts with a square and proceeds as follows

- each square is replaced by 5 smaller squares each $\frac{1}{9}$ th the size of the square being replaced
- the 5 smaller squares are the ones in each corner and the one in the centre
- once each of the squares has been replaced, the square immediately to the right and above the centre square of the pattern is then removed

Let u_n be the number of squares in the pattern in stage n , where stage 1 is the original square.

(a) Explain why u_n satisfies the recurrence system

$$u_1 = 1 \quad u_{n+1} = 5u_n - 1 \quad (n = 1, 2, 3, \dots) \quad (2)$$

(b) Solve this recurrence system.

(5)

Given that the initial square has area 25

(c) determine the total area of all the squares in stage 8 of the pattern, giving your answer to 2 significant figures.

(2)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



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Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/22**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

22: Further Pure Mathematics 2

(Part of option A only)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ►

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1. The operation $*$ is defined on the set $G = \{0, 1, 2, 3\}$ by

$$x * y \equiv x + y - 2xy \pmod{4}$$

- (a) Complete the Cayley table below.

(2)

$*$	0	1	2	3
0				
1				
2				
3				

- (b) Show that G is a group under the operation $*$

(You may assume the associative law is satisfied.)

(3)

- (c) State the order of each element of G .

(2)

- (d) State whether G is a cyclic group, giving a reason for your answer.

(1)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is represented by the matrix

$$\mathbf{M} = \begin{pmatrix} 5 & 1 \\ k & -3 \end{pmatrix}$$

where k is a constant.

Given that matrix $\mathbf{M}$ has a repeated eigenvalue,

(a) determine

- (i) the value of k
- (ii) the eigenvalue.

(6)

(b) Hence determine a Cartesian equation of the invariant line under T .

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

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(Total for Question 2 is 8 marks)



3. A complex number z is represented by the point P on an Argand diagram.

Given that

$$\arg\left(\frac{z - 4 - i}{z - 2 - 7i}\right) = \frac{\pi}{2}$$

- (a) sketch the locus of P as z varies, (2)
- (b) determine the exact maximum possible value of $|z|$ (5)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



4. A student takes out a loan for £1000 from a bank.

The bank charges 0.5% monthly interest on the amount of the loan yet to be repaid.

At the end of each month

- the interest is added to the loan
- the student then repays £50

Let U_n be the amount of money owed n months after the loan was taken out.

The amount of money owed by the student is modelled by the recurrence relation

$$U_n = 1.005U_{n-1} - A \quad U_0 = 1000 \quad n \in \mathbb{Z}^+$$

where A is a constant.

- (a) (i) State the value of the constant A .

- (ii) Explain, in the context of the problem, the value 1.005

(2)

Using the value of A found in part (a)(i),

- (b) solve the recurrence relation

$$U_n = 1.005U_{n-1} - A \quad U_0 = 1000 \quad n \in \mathbb{Z}^+$$

(5)

- (c) Hence determine, according to the model, the number of months it will take to completely repay the loan.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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(Total for Question 4 is 9 marks)



5. (i) Making your reasoning clear and using modulo arithmetic, show that

214^6 is divisible by 8

(3)

- (ii) The following 7-digit number has four unknown digits

$\boxed{a}5\boxed{b}8\boxed{a}\boxed{b}0$

Given that the number is divisible by 11

- (a) determine the value of the digit a .

(2)

Given that the number is also divisible by 3

- (b) determine the possible values of the digit b .

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 8 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

8FM0/22

Further Mathematics

Advanced Subsidiary

Further Mathematics options

22: Further Pure Mathematics 2

(Part of option A only)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ►

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Pearson

1. Sketch on an Argand diagram the region defined by

$$\left\{z \in \mathbb{C} : -\frac{\pi}{4} < \arg(z + 2) < \frac{\pi}{4}\right\} \cap \{z \in \mathbb{C} : -1 < \operatorname{Re}(z) \leq 1\}$$

On your sketch

- shade the part of the diagram that is included in the region
- use solid lines to show the parts of the boundary that are included in the region
- use dashed lines to show the parts of the boundary that are not included in the region

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 4 marks)



2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$\mathbf{M} = \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix}$$

Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{D}$$

(7)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



- (3)

- | | | | | | | | | |
|---------------|----|----|----|----|----|----|----|----|
| $\times_{30}$ | 2 | 4 | 8 | 14 | 16 | 22 | 26 | 28 |
| 2 | 4 | 8 | 16 | 28 | 2 | 14 | 22 | 26 |
| 4 | 8 | | 2 | | | 28 | 14 | |
| 8 | 16 | 2 | | | 8 | | | 14 |
| 14 | 28 | | 22 | 16 | | 8 | 4 | |
| 16 | 2 | 4 | | 14 | 16 | | | |
| 22 | 14 | | 26 | | | 4 | 2 | 16 |
| 26 | 22 | 14 | | 4 | | | | 8 |
| 28 | 26 | | 14 | | 28 | | 8 | |

(b) Hence determine whether the set X with the operation of multiplication modulo 30 forms a group.

(6)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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$\times_{30}$	2	4	8	14	16	22	26	28
2	4	8	16	28	2	14	22	26
4	8		2			28	14	
8	16	2			8			14
14	28		22	16		8	4	
16	2	4		14	16			
22	14		26			4	2	16
26	22	14		4				8
28	26		14		28		8	



4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) (a) Use the Euclidean algorithm to find the highest common factor h of 416 and 72

- (b) Hence determine integers a and b such that

$$416a + 72b = h \tag{3}$$

- (c) Determine the value c in the set $\{0,1,2,\dots,415\}$ such that

$$23 \times 72 \equiv c \pmod{416} \tag{2}$$

- (ii) Evaluate $5^{10} \pmod{13}$ giving your answer as the smallest positive integer solution. (3)



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Question 4 continued

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DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 11 marks)



5. A person takes a course of a particular vitamin.

Before the course there was none of the vitamin in the person's body.

During the course, vitamin tablets are taken at the same time each day.

Initially two tablets are taken and on each following day only one tablet is taken.

Each tablet contains 10 mg of the vitamin.

Between doses the amount of the vitamin in the person's body decreases naturally by 60%

Let u_n mg be the amount of the vitamin in the person's body immediately after a tablet is taken, n days after the initial two tablets were taken.

- (a) Explain why u_n satisfies the recurrence relation

$$u_0 = 20 \quad u_{n+1} = 0.4u_n + 10 \quad (2)$$

The general solution to this recurrence relation has the form $u_n = a(0.4)^n + b$

- (b) Determine the value of a and the value of b . (4)

The course is only effective if the amount of the vitamin in the person's body remains above 6 mg at **all times** throughout the course.

- (c) Determine whether this course of the vitamin will be effective for this person, giving a reason for your answer. (3)



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Question 5 continued

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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Thursday 08 October 2020			
Afternoon		Paper Reference 8FM0/22	
Further Mathematics Advanced Subsidiary Further Mathematics options 22: Further Pure Mathematics 2 (Part of option A only)			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

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Turn over ►

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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. The highest common factor of 963 and 657 is c .

(a) Use the Euclidean algorithm to find the value of c .

(3)

(b) Hence find integers a and b such that

$$963a + 657b = c$$

(3)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3. (i)

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 1 & 4 \end{pmatrix}$$

(a) Show that the characteristic equation for $\mathbf{A}$ is $\lambda^2 - 5\lambda + 6 = 0$ (2)

(b) Use the Cayley-Hamilton theorem to find integers p and q such that

$$\mathbf{A}^3 = p\mathbf{A} + q\mathbf{I} \quad (3)$$

(ii) Given that the 2×2 matrix $\mathbf{M}$ has eigenvalues $-1 + i$ and $-1 - i$,
with eigenvectors $\begin{pmatrix} 1 \\ 2 - i \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 + i \end{pmatrix}$ respectively, find the matrix $\mathbf{M}$. (5)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4. Sam borrows £10 000 from a bank to pay for an extension to his house. The bank charges 5% annual interest on the portion of the loan yet to be repaid. Immediately after the interest has been added at the end of each year and before the start of the next year, Sam pays the bank a fixed amount, £ F .

Given that £ A_n (where $A_n \geq 0$) is the amount owed at the start of year n ,

- (a) write down an expression for A_{n+1} in terms of A_n and F , (1)

- (b) prove, by induction that, for $n \geq 1$

$$A_n = (10\,000 - 20F)1.05^{n-1} + 20F$$
(5)

- (c) Find the smallest value of F for which Sam can repay all of the loan by the start of year 16. (4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 10 marks)



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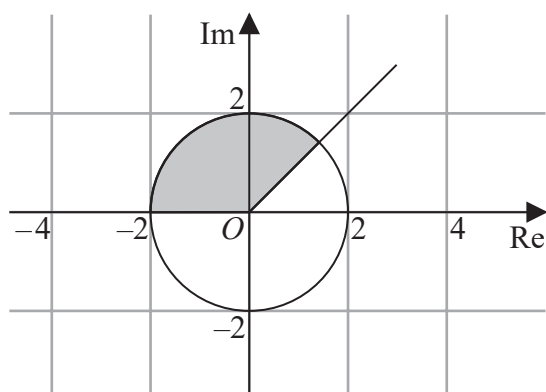


Figure 1

Figure 1 shows an Argand diagram.

The set of points, A , that lies within the shaded region, including its boundaries, is defined by

$$A = \{z : p \leq \arg(z) \leq q\} \cap \{z : |z| \leq r\}$$

where p , q and r are positive constants.

(a) Write down the values of p , q and r .

(2)

Given that $w = -2\sqrt{3} + 2i$ and $z \in A$,

(b) find the maximum value of $|w - z|^2$ giving your answer in an exact simplified form.

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 6 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Thursday 16 May 2019			
Afternoon		Paper Reference 8FM0-22	
Further Mathematics Advanced Subsidiary Further Mathematics options 22: Further Pure Mathematics 2 (Part of option A only)			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. (i) Determine all the possible integers a , where $a > 3$, such that

$$15 \equiv 3 \pmod{a} \quad (2)$$

- (ii) Show that if p is prime, x is an integer and $x^2 \equiv 1 \pmod{p}$ then either

$$x \equiv 1 \pmod{p} \quad \text{or} \quad x \equiv -1 \pmod{p} \quad (3)$$

- (iii) A company has £13 940 220 to share between 11 charities.

Without performing any division and showing all your working, decide if it is possible to share this money equally between the 11 charities.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. A curve C in the complex plane is described by the equation

$$|z - 1 - 8i| = 3|z - 1|$$

- (a) Show that C is a circle, and find its centre and radius.

(4)

- (b) Using the answer to part (a), determine whether $z = 3 - 3i$ satisfies the inequality

$$|z - 1 - 8i| \geq 3|z - 1|$$

(2)

- (c) Shade, on an Argand diagram, the set of points that satisfies both

$$|z - 1 - 8i| \geq 3|z - 1| \quad \text{and} \quad 0 \leq \arg(z + i) \leq \frac{\pi}{4}$$

(4)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4. The set $\{e, p, q, r, s\}$ forms a group, A , under the operation $*$

Given that e is the identity element and that

$$p * p = s \quad s * s = r \quad p * p * p = q$$

- (a) show that

(i) $p * q = r$

(ii) $s * p = q$

(2)

- (b) Hence complete the Cayley table below.

$*$	e	p	q	r	s
e					
p					
q					
r					
s					

A spare table can be found on page 11 if you need to rewrite your Cayley table.

(2)

- (c) Use your table to find $p * q * r * s$

(1)

A student states that there is a subgroup of A of order 3

- (d) Comment on the validity of this statement, giving a reason for your answer.

(2)



This image shows a full page of blank, lined paper. It features approximately 20 horizontal grey lines spaced evenly apart, typical of notebook paper. The lines extend across the entire width of the page, leaving small margins at the top and bottom. There are no vertical lines or other markings present.

$*$	e	p	q	r	s
e					
p					
q					
r					
s					

5. On Jim's 11th birthday his parents invest £1000 for him in a savings account.

The account earns 2% interest each year.

On each subsequent birthday, Jim's parents add another £500 to this savings account.

Let U_n be the amount of money that Jim has in his savings account n years after his 11th birthday, once the interest for the previous year has been paid and the £500 has been added.

- (a) Explain, in the context of the problem, why the amount of money that Jim has in his savings account can be modelled by the recurrence relation of the form

$$U_n = 1.02U_{n-1} + 500 \qquad U_0 = 1000 \qquad n \in \mathbb{Z}^+ \qquad (3)$$

- (b) State an assumption that must be made for this model to be valid. (1)

- (c) Solve the recurrence relation

$$U_n = 1.02U_{n-1} + 500 \qquad U_0 = 1000 \qquad n \in \mathbb{Z}^+ \qquad (5)$$

Jim hopes to be able to buy a car on his 18th birthday.

- (d) Use the answer to part (c) to find out whether Jim will have enough money in his savings account to buy a car that costs £4 500 (2)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 11 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



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Level 3 GCE

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Further Mathematics

Advanced Subsidiary

Further Mathematics options

22: Further Pure Mathematics 2

(Part of option A only)

Thursday 17 May 2018 – Afternoon

Paper Reference

8FM0/22

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B). Coloured pencils and highlighter pens must not be used.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- When a calculator is used, the answer should be given to an appropriate degree of accuracy.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. (i) Using a suitable algorithm and without performing any division, determine whether 23 738 is divisible by 11

(2)

- (ii) Use the Euclidean algorithm to find the highest common factor of 2322 and 654

(3)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2.

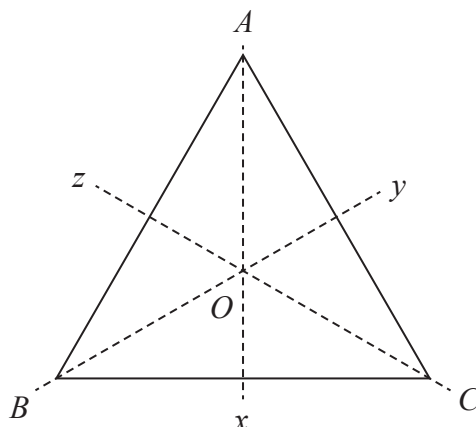


Figure 1

Figure 1 shows an equilateral triangle ABC . The lines x , y and z and their point of intersection, O , are fixed in the plane. The triangle ABC is transformed about these fixed lines and the fixed point O . The lines x , y and z each pass through a vertex of the triangle and the midpoint of the opposite side.

The transformations I , X , Y , Z , R_1 and R_2 of the plane containing triangle ABC are defined as follows:

- I : Do nothing
- X : Reflect in the line x
- Y : Reflect in the line y
- Z : Reflect in the line z
- R_1 : Rotate 120° anticlockwise about O
- R_2 : Rotate 240° anticlockwise about O

The operation $*$ is defined as 'followed by' on the set $T = \{I, X, Y, Z, R_1, R_2\}$.

For example, $X * Y$ means a reflection in the line x followed by a reflection in the line y .

(a) (i) Complete the Cayley table on page 5

Given that the associative law is satisfied,

(ii) show that T is a group under the operation $*$

(6)

(b) Show that the element R_2 has order 3

(2)

(c) Explain why T is not a cyclic group.

(1)

(d) Write down the elements of a subgroup of T that has order 3

(1)



		Second transformation					
	*	I	X	Y	Z	R_1	R_2
First Transformation	I						
	X		I			Z	
	Y						
	Z						
	R_1		Y				
	R_2						

Question 2 continued

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[illegible]

- 3** A tree at the bottom of a garden needs to be reduced in height. The tree is known to increase in height by 15 centimetres each year.

On the first day of every year, the height is measured and the tree is immediately trimmed by 3% of this height.

When the tree is measured, before trimming on the first day of year 1, the height is 6 metres.

Let H_n be the height of the tree immediately before trimming on the first day of year n .

- (a) Explain, in the context of the problem, why the height of the tree may be modelled by the recurrence relation

$$H_{n+1} = 0.97H_n + 0.15, \quad H_1 = 6, \quad n \in \mathbb{Z}^+ \quad (3)$$

- (b) Prove by induction that $H_n = 0.97^{n-1} + 5$, $n \geq 1$

- (c) Explain what will happen to the height of the tree immediately before trimming in the long term.

- (d) By what fixed percentage should the tree be trimmed each year if the height of the tree immediately before trimming is to be 4 metres in the long term? (2)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$$

Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

(7)

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5. A complex number z is represented by the point P on an Argand diagram.

Given that $\arg\left(\frac{z - 6i}{z - 3i}\right) = \frac{\pi}{3}$

- (a) sketch the locus of P as z varies, (3)

- (b) find the exact maximum possible value of $|z|$ (5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 8 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS



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Other names

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Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced Subsidiary

Further Mathematics options

Paper 2A: Further Pure Mathematics 1 and Further Pure Mathematics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 40 minutes

Paper Reference

8FM0/2A

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

6. Given that

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 6 & 4 \end{pmatrix}$$

(a) find the characteristic equation of the matrix $\mathbf{A}$.

(2)

(b) Hence show that $\mathbf{A}^3 = 43\mathbf{A} - 42\mathbf{I}$.

(3)

Question 6 continued

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(Total for Question 6 is 5 marks)

7. (i) Without performing any division, explain why 8184 is divisible by 6

(2)

- (ii) Use the Euclidean algorithm to find integers a and b such that

$$27a + 31b = 1$$

(4)

Question 7 continued

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(Total for Question 7 is 6 marks)

8. A curve C is described by the equation

$$|z - 9 + 12i| = 2|z|$$

(a) Show that C is a circle, and find its centre and radius.

(4)

(b) Sketch C on an Argand diagram.

(2)

Given that w lies on C ,

(c) find the largest value of a and the smallest value of b that must satisfy

$$a \leq \operatorname{Re}(w) \leq b$$

(2)

Question 8 continued

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(Total for Question 8 is 8 marks)

9. The operation $*$ is defined on the set $S = \{0, 2, 3, 4, 5, 6\}$ by $x*y = x + y = xy \pmod{7}$

$*$	0	2	3	4	5	6
0						
2		0				
3						5
4						
5		4				
6						

- (a) (i) Complete the Cayley table shown above
- (ii) Show that S is a group under the operation $*$
- (You may assume the associative law is satisfied.)
- (6)
- (b) Show that the element 4 has order 3
- (2)
- (c) Find an element which generates the group and express each of the elements in terms of this generator.
- (3)

Question 9 continued

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(Total for Question 9 is 11 marks)

Question 10 continued

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Question 10 continued

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(Total for Question 10 is 10 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names
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Pearson Edexcel

Level 3 GCE

Centre Number

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Candidate Number

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Specimen Paper

Paper Reference **8FM0/22**

Further Mathematics
Advanced Subsidiary
22: Further Pure Mathematics 2

You must have:
 Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 4 questions in this question paper. The total mark for this paper is 40.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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(3)

(1)

(1)

- (3)

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Question 1 continued

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Only use this grid if you need to rewrite the Cayley table.

$\times_{13}$	1	3	4	9	10	12
1						
3						
4						
9						
10						
12						

(Total for Question 1 is 8 marks)



2. The two sequences x_n and y_n satisfy the recurrence relations

$$\begin{aligned} x_1 &= 2 & x_{n+1} &= 5x_n - y_n \\ y_1 &= 1 & y_{n+1} &= 3x_n + y_n \end{aligned} \quad n \geq 1$$

(a) Find the value of x_3

(2)

These recurrence relations can be written in matrix form as

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x_n \\ y_n \end{pmatrix} \quad n \geq 1$$

where $\mathbf{A} = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$

(b) Find

(i) the eigenvalues of $\mathbf{A}$,

(ii) the corresponding eigenvectors of $\mathbf{A}$.

(5)

(c) Hence write down matrices $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$, where $\mathbf{D}$ is a diagonal matrix.

(2)

New sequences u_n and v_n can be formed from x_n and y_n using the transformation

$$\begin{pmatrix} u_n \\ v_n \end{pmatrix} = \mathbf{P}^{-1} \begin{pmatrix} x_n \\ y_n \end{pmatrix}$$

(d) Show that $\begin{pmatrix} u_{n+1} \\ v_{n+1} \end{pmatrix} = \mathbf{D} \begin{pmatrix} u_n \\ v_n \end{pmatrix}$

(2)

(e) Hence find closed form expressions for the original sequences x_n and y_n

(5)



Question 2 continued

Lined area for writing the answer to Question 2 continued.

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Question 2 continued

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Question 2 continued

(Total for Question 2 is 16 marks)



3. (a) Sketch, on an Argand diagram, the arc given by the locus of points z in the complex plane satisfying

$$\arg\left(\frac{z}{z-25i}\right) = \frac{11\pi}{12} \quad (2)$$

The centre of the circle containing this arc is $x + yi$ where

$$x = -\frac{25}{2}(2 + \sqrt{3})$$

- (b) Find the radius of this circle, giving your answer to 3 decimal places. (4)

In a game, players take turns to roll balls along a horizontal surface from a fixed starting position to a fixed target point 25 metres away.

The game can be modelled in the complex plane, with the starting position at the origin and the target at the point $25i$, where the units are metres.

In a particular game, the first player's ball has come to rest with its centre at $0.5 + 23.25i$. The second player's ball is following the path sketched in part (a).

Given that the radius of each ball is approximately 5 centimetres,

- (c) use this model to determine whether the second player's ball will hit the first player's ball.



Question 3 continued

Lined area for writing the answer to Question 3 continued.

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Question 3 continued

Lined area for writing the answer to Question 3.

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Question 3 continued

(Total for Question 3 is 10 marks)



4. A positive integer n satisfies the inequality $99 < n < 1000$
The integer n is written as a three digit number abc such that

$$n = 100a + 10b + c$$

where the digits a , b , and c are each integers between 0 and 9 inclusive, $a \neq 0$

The integer n has the following three properties:

- n is divisible by 11
- the sum of the digits of n is odd
- $n \equiv 5 \pmod{9}$

Use this information to find all possible values for n , making your reasoning clear.

(6)

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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



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Question 4 continued

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(Total for Question 4 is 6 marks)

TOTAL FOR PAPER IS 40 MARKS

